

**RAMCO INSTITUTE OF TECHNOLOGY**  
**Department of Computer Science and Engineering**  
**Academic Year: 2019 - 2020 (Odd Semester)**

**Degree, Semester & Branch** : V Semester B.E (Common to ECE, EEE, MECH)  
**Course(Subject) Code & Title** : OIT551 Database Management Systems  
**Name of the Faculty member** : Mr. K.Vignesh Saravanan, AP / CSE

**Date:** 27.08.2019

**Innovative Practice: Collaborative Coding**  
**Topics: ODBC/JDBC connectivity, Embedded SQL**

**Objectives:**

- **O1:** Analyze, think and work as a team to develop the code (correct the code) for database connectivity in C / JAVA language.

**Justification for choosing this topic:**

Developing code is an integral part across all various domains. Embedded SQL and ODBC/JDBC connectivity is a standard application programming interface for accessing database management systems. The prime objective of Collaborative coding is to make the students work as a team and develop the code for database connectivity.

**Activity on Collaborative Coding:**

After completion of the unit / topics this activity can be planned as a team activity, in order to make the students understand the program codes very clearly and to improve their level of understanding and gain more knowledge in coding and logical problem solving. The students are provided with materials (like worksheet) with partial codes and some of the program codes are shuffled. The task of the students is to fill the correct statements and complete the partial codes and also rearrange the shuffled codes in correct order and complete the program. The course instructor will provide sufficient hints and correct syntax and will discuss all the doubts in the same class session and make them understand and complete the coding. The course instructor also provided the correct answers and explained how the code gets executed in the end of the session.

**Observations made:**

- Some prerequisite on JAVA syntax and coding should be given to some students.
- Some of the students could understand the working principle of the ODBC and JDBC connectivity, but felt very difficult to code due to lack of coding knowledge.

**OUTCOME:**

- Through this activity the students can deeply **understand and write codes to establish the database connectivity** using MySQL and Oracle software.
- The **students are made to write and rearrange the program codes** mainly the connection establishment codes which gave them complete understanding over the steps involved in database connectivity.

## Screenshots of the activity done by the students

```
#include <stdio.h>
EXEC SQL INCLUDE ...SQLCA...;
main()
{
EXEC SQL BEGIN DECLARE SECTION;
    int marks = 90;
    int studno;
    char name[50];
    char grade[10];
EXEC SQL END DECLARE SECTION;
EXEC SQL CONNECT ...;
EXEC SQL WHENEVER SQLERROR GOTO error;
EXEC SQL DECLARE ...CURSOR FOR
    select ...marks, name, grade from ...student
    where ...marks > 90;
EXEC SQL OPEN ...stud;
For(...)
{
EXEC SQL FETCH ...stud INTO
    ...marks, ...studno, ...name;
    printf("Student no. is: %d", studno);
    printf("Name is: %s", name);
    printf("Grade is: %s", grade);
}
Error:
    printf("SQL code error");
EXEC SQL CLOSE ...stud;
```

Consider the relation student (studno, mark, name, and grade). Write embedded, statements to retrieve all the students records whose mark is more than 90

```
#include <sqlca.h>
EXEC SQL INCLUDE ...SQLCA...;
main()
{
EXEC SQL BEGIN DECLARE SECTION;
    int mark = 90;
    int studno;
    char hname;
    char hgrade;
EXEC SQL END DECLARE SECTION;
EXEC SQL CONNECT ...;
EXEC SQL WHENEVER SQLERROR GOTO error;
EXEC SQL DECLARE ...CURSOR FOR
    select :studno, :hname, :hgrade from ...student
    where ...mark > 90;
EXEC SQL CLOSE ...cursor;
For(...)
{
EXEC SQL FETCH ...cursor INTO
    :studno, :hname, :hgrade;
    printf("studno = %d", studno);
    printf("name = %s", hname);
    printf("grade = %s", hgrade);
}
Error:
    printf("SQL code error");
EXEC SQL CLOSE ...cursor;
```

Program to print the regno, names of all students scored above n

|                                                    |    |   |    |   |
|----------------------------------------------------|----|---|----|---|
| For(;;) {                                          | 16 | X | 8  | 8 |
| {                                                  |    |   | 4  |   |
| Error:                                             |    |   |    |   |
| EXEC SQL CONNECT .....                             | 10 | X | 5  | 6 |
| int h_mark;                                        | 6  | ✓ |    |   |
| Printf("Regno = %d", h_regno);                     | 19 | X | 11 |   |
| printf("Enter the mark");                          | 20 | X | 12 |   |
| EXEC SQL INCLUDE SQLCA;                            | 2  | ✓ | 2  |   |
| select regno, name from student                    | 13 | X |    |   |
| main()                                             | 3  | ✓ | 7  |   |
| int h_regno;                                       | 7  | ✓ |    |   |
| }                                                  | 22 | ✓ |    |   |
| EXEC SQL DECLARE cursor_mark CURSOR FOR            | 12 | X | 8  |   |
| EXEC SQL BEGIN DECLARE SECTION;                    | 5  | ✓ | 5  |   |
| #include<stdio.h>                                  | 1  | ✓ | 10 |   |
| Scanf("%d", &h_mark);                              | 18 | X |    |   |
| where mark= :h_mark;                               | 14 | X |    |   |
| EXEC SQL OPEN cursor_mark; {                       | 15 | X |    |   |
| Printf("SQL code error");                          | 24 | ✓ |    |   |
| EXEC SQL CLOSE cursor_mark;                        | 25 | ✓ |    |   |
| EXEC SQL FETCH cursor_mark INTO :h_regno, :h_name; | 17 | X | 16 |   |
| Printf("Name = %s", h_name);                       | 31 | ✓ |    |   |
| char h_name[50];                                   | 3  | ✓ |    |   |
| EXEC SQL END DECLARE SECTION;                      | 9  | ✓ |    |   |
| EXEC SQL WHENEVER SQLERROR GOTO error;             | 11 | X | 6  | 7 |

Program that asks the user for a person number and print out all their properties (propertyNo, street, city)

|                                               |    |   |
|-----------------------------------------------|----|---|
| Error:                                        | 18 | ✓ |
| Printf("SQL code error");                     | 14 | ✓ |
| For(;;) {                                     |    |   |
| int h_personID;                               | 5  | ✓ |
| char h_propertyNo;                            |    |   |
| char h_street[50];                            | 6  | ✓ |
| char h_city[20];                              |    |   |
| EXEC SQL CLOSE cursor_property;               |    | X |
| EXEC SQL CONNECT :username;                   | 8  | X |
| printf("Enter the personID");                 |    | X |
| Scanf("%d", &h_personID);                     | 9  | X |
| EXEC SQL DECLARE cursor_property CURSOR FOR   |    |   |
| #include<stdio.h>                             | 1  | ✓ |
| EXEC SQL INCLUDE SQLCA;                       | 2  | ✓ |
| EXEC SQL FETCH cursor_property INTO           |    | X |
| : h_propertyNo, : h_street, : h_city;         |    |   |
| EXEC SQL OPEN cursor_property;                |    | X |
| }                                             |    | X |
| main() {                                      | 2  | ✓ |
| EXEC SQL BEGIN DECLARE SECTION;               | 4  | ✓ |
| select proertyNo, street, city from property  | 12 | ✓ |
| where mark= :h_mark;                          |    |   |
| Printf("Property number = %d", h_propertyNo); |    |   |
| Printf("Street = %s", h_street);              | 11 | X |
| Printf("City = %s", h_city);                  |    |   |
| EXEC SQL WHENEVER SQLERROR GOTO error;        | 10 | ✓ |
| EXEC SQL END DECLARE SECTION;                 | 7  | ✓ |

Relevance Mapping - This activity helps in attaining the following Objectives – PO mapping:

| <b>Course Outcome /<br/>Programme Outcome</b>                                                                                | <b>PO1</b> | <b>PO2</b> | <b>PO3</b> | <b>PO9</b> | <b>PO10</b> | <b>PO12</b> |
|------------------------------------------------------------------------------------------------------------------------------|------------|------------|------------|------------|-------------|-------------|
| O1 - Analyse, think and work as a team to develop the code (correct the code) for database connectivity in C / JAVA language | <b>3</b>   | <b>3</b>   | <b>3</b>   | <b>2</b>   | <b>1</b>    | <b>1</b>    |

### **Reflection Critique**

#### **Challenges:**

- Some of the students did not show interest in coding with C, Java and DB connection.
- Slow learners were not able to easily solve, even the basic coding part. So the students felt very difficult in this activity and could not complete in time.
- Individual and Team assessment/observation were difficult.

#### **Steps taken for address the challenges:**

- I motivated the students explaining with some applications relevant to their domain and made aware that coding is an integral part of all applications.
- I provided more hints and help to make the students solve the coding part.

#### **References:**

1. <https://teaching.cornell.edu/teaching-resources/engaging-students/collaborative-learning>
2. <https://www.ritrjpm.ac.in/images/computer-science/collaborative-tutorial-sm.pdf>

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